

STATISTICS 5.2. Permutations + Combinations

1. Factorials

"five factorial"

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1$$

$$0! = 1$$

integer; $n > 0$

Ex. 1.: $\frac{5!}{3!} = \frac{5 \times 4 \times \cancel{3!}}{\cancel{3!}} = 5 \times 4 = 20$

Ex. 2.:

M A T H E M A T I C A L is

* with repeats

12 letters

$$\frac{12!}{2! \times 2! \times 3!} = 19958400$$

Two Ms repeated

Two Ts repeated

Three As repeated

2. Permutations

- the order matters, arrangement matters



I should be one of the judges! When I heard the 16 songs in the competition, I knew which ones I thought were the best three. Last night they announced the results and I had picked the same three songs in the same order as the judges!

The winner can be chosen in 16 ways

2nd song 15

3rd song 14

→ total # of ways: $16 \times 15 \times 14 = 3360$

→ probability of Joyeeta picking the right way $= \frac{1}{3360}$

permutation of 3 objects from 16

$${}^{16}P_3 = 16 \times 15 \times 14$$

$$= \frac{16 \times 15 \times 14 \times \cancel{13!}}{\cancel{13!}}$$

$$= \frac{16 \times 15 \times 14 \times \cancel{13!}}{(16-3)!}$$

total objects

chosen objects

$${}^nP_r = \frac{n!}{(n-r)!}$$

3. Combinations - order DOESN'T matter

A maths teacher is playing a game with her students. Each student selects **six numbers** out of a **possible 49** (numbers 1, 2, ..., 49). The maths teacher then uses a **random number machine** to generate six numbers. If a student's numbers **match** the teacher's numbers then they **win** a prize.

- You have the six winning numbers. Does it matter in which order the machine picked them?

If the order mattered: ${}^{49}P_6 = 49 \times 48 \times 47 \times 46 \times 45 \times 44$

but the order doesn't matter, so picking

1, 8, 16, 15, 24, 47

or 47, 1, 8, 16, 24, 15

.. are **all the same!**

there are **6!** ways of arranging these

$$\# \text{ of ways} = \frac{49 \times 48 \times 47 \times 46 \times 45 \times 44}{6!}$$

$$= \frac{{}^{49}P_6}{6!} = \frac{49!}{6! \times (49-6)!}$$

total objects →

chosen objects →

selected from →

selected →

not selected →

$${}_r^n C = \frac{n!}{r!(n-r)!} = \frac{{}_r^n P}{r!}$$

4. Finding probabilities of certain events

$$P(\text{event}) = \frac{\text{No. favourable permutations}}{\text{No. possible permutations}}$$

$$P(\text{event}) = \frac{\text{No. favourable combinations}}{\text{No. possible combinations}}$$